Fan Huang

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Luddy School of Informatics, Computing, and Engineering
Indiana University Bloomington

Education

2023–*	**Ph.D.** in Informatics (Complex Networks and Systems), Indiana University Bloomington, USA Advisors: Prof. Jisun An and Prof. Haewoon Kwak
2022–23	**Ph.D.** in Computer Science, Singapore Management University, Singapore
2020–22	**M.Sc.** in Information Systems, Nanyang Technological University, Singapore Thesis: *Public Opinion Trend Analysis on Nuclear Energy via Aspect-based Sentiment Analysis Methods*
2014–18	**B.Sc.** in Computer Science & Technology, Central South University, China

Research Experience

2023–	Research Assistant, Indiana University Bloomington Advisors: Prof. Jisun An and Prof. Haewoon Kwak Collaborators: Prof. Yong-Yeol Ahn, Prof. Filippo Menczer, and Prof. Filippo Radicchi
2024	Quantitative Research Intern, Core Research Team, QuantLab Financial, Houston (Jun–Aug 2024)
2024	Research Summer Internship, Indiana University Bloomington, supervised by Prof. Katy Börner (May–Jun 2024)
2022–23	Research Assistant, Singapore Management University

Publications

Google Scholar
† → Equal contribution

Preprints / Under Review

J1. **Huang, Fan**. Catalyst Papers in Artificial Intelligence Research: A Landscape on ICLR from 2017 to 2025. *arXiv preprint* (2026).

J2. **Huang, Fan**. Reasoning Topology Matters: Network-of-Thought for Complex Reasoning Tasks. *arXiv preprint arXiv:2603.20730* (2026).

J3. **Huang, Fan**. ReFlect: An Effective Harness System for Complex Long-Horizon LLM Reasoning. *arXiv preprint* (2026).

*Expected 2027.

J4. **Huang, Fan**. Simulating Hate Speech Cascades with Multi-LLM Agents: Empirical Grounding, Modeling Fidelity, and Intervention Strategies. *arXiv preprint* (2026).

J5. **Huang, Fan** & Kim, M. LLMs as Implicit Probabilistic World Models: Distributional Stability, Convergence, and Scale Discordance. *OpenReview*. https://openreview.net/pdf?id=eBNpytJp2L (2026).

J6. **Huang, Fan** & Kim, M. Can LLMs Predict Academic Collaboration? Topology Heuristics vs. LLM-Based Link Prediction on Real Co-authorship Networks. *arXiv preprint arXiv:2604.01379* (2026).

J7. **Huang, Fan**, Tong, D., Chen, C., Kwak, H. & An, J. Instruction-Conditioned Motion Control (ICMC) for Robotic Arms: VLA Decomposition and Reinforcement Learning Tuning. *OpenReview* (2026).

J8. **Huang, Fan**, Zhang, S., Kwak, H. & An, J. CogBias: Measuring and Mitigating Cognitive Bias in Large Language Models. *arXiv preprint arXiv:2604.01366* (2026).

J9. **Huang, Fan** & Zhou, H. Comparative Evaluation of LLM-Based Review Paradigms for Academic Paper Quality Assessment. *OpenReview* (2026).

J10. Qi, W., **Huang, Fan**, An, J. & Kwak, H. A Cross-Cultural Comparison of LLM-based Public Opinion Simulation: Evaluating Chinese and US Models on Diverse Societies. *arXiv preprint arXiv:2506.21587* (2025).

J11. Seckin, O. C., Nascimento Silva, F., Truong, B. T., Kim, S., **Huang, Fan**, Liu, N., Flammini, A. & Menczer, F. The Rise of Bluesky. *arXiv preprint* (2025).

J12. Yin, L. & **Huang, Fan**. Dif: A Framework for Benchmarking and Verifying Implicit Bias in LLMs. *arXiv preprint arXiv:2505.10013* (2025).

J13. **Huang, Fan**, Kwak, H. & An, J. Token-Ensemble Text Generation: On Attacking the Automatic AI-Generated Text Detection. *arXiv preprint* (2024).

J14. **Huang, Fan**. Highlight Timestamp Detection Model for Comedy Videos via Multimodal Sentiment Analysis. *arXiv preprint*. https://arxiv.org/abs/2106.00451 (2021).

J15. **Huang, Fan**. Network Activities Recognition and Analysis Based on Supervised Machine Learning Classification Methods Using J48 and Naïve Bayes Algorithm. *arXiv preprint*. https://arxiv.org/abs/2105.13698 (2021).

Peer-reviewed Conference Proceedings

C1. **Huang, Fan**. *LLM Belief Atlas* in *InfoVis 2026 (Poster)* (2026).

C2. **Huang, Fan**, Kwak, H. & An, J. *Understanding Moral Reasoning Trajectories in Large Language Models: Toward Probing-Based Explainability* in *Findings of the Association for Computational Linguistics (EMNLP Findings 2026)* (2026).

C3. **Huang, Fan**, Kwak, H. & An, J. *Vulnerability of LLMs' Stated Belief? LLMs Belief Resistance Check Through Strategic Persuasive Conversation Interventions* in *Findings of the Association for Computational Linguistics (ACL Findings 2026)* (2026).

C4. Qi, W., **Huang, Fan**, Muralidharan, R., An, J. & Kwak, H. *XChoice: Explainable Evaluation of AI-Human Alignment in LLM-based Constrained Choice Decision Making* in *Findings of the Association for Computational Linguistics (AACL-IJCNLP Findings 2026)* (2026).

C5. Qi, W., **Huang, Fan**, An, J. & Kwak, H. *Is DeepSeek a New Voice Among LLMs in Public Opinion Simulation?* in *Proceedings of the IEEE International Conference on Data Mining (ICDM 2025)* (2025).

C6. **Huang, Fan**, Kwak, H., Park, K. & An, J. *ChatGPT Rates Like Human: Towards a Better Alignment of Text Explanation Quality Assessments* in *Proceedings of the Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (2024).

C7. **Huang, Fan**, Kwak, H. & An, J. *Chain of Explanation: New Prompting Method to Generate Higher Quality Natural Language Explanation for Implicit Hate Speech* in *Companion Proceedings of the Web Conference (WWW 2023)* (2023).

C8. **Huang, Fan**, Kwak, H. & An, J. *Is ChatGPT Better than Human Annotators? Potential and Limitations of ChatGPT in Explaining Implicit Hate Speech* in *Companion Proceedings of the Web Conference (WWW 2023)* (2023).

C9. Liaw, S., **Huang, Fan**, Benevenuto, F., Kwak, H. & An, J. *YouNICon: YouTube's CommuNIty of Conspiracy Videos* in *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2023)* (2023).

Teaching

Indiana University Bloomington – Luddy School of Informatics, Computing, and Engineering

2026	Teaching Assistant, INFO-I 201 Mathematical Foundations of Informatics (Fall 2026)
2026	Lecture Presentation, Vision Language Models (VLM), INFO-I 590 Foundation and Application for Large Language Models (Apr 2026)
2026	Lecture Presentation, Retrieval Augmented Generation (RAG), INFO-I 590 Foundation and Application for Large Language Models (Mar 2026)
2026	Teaching Assistant, INFO-I 590 Foundation and Application for Large Language Models (Spring 2026)
2025	Teaching Assistant, INFO-I 590 Foundation and Application for Large Language Models (Fall 2025)
2025	Teaching Assistant, INFO-I 590 Foundation and Application for Large Language Models (Spring 2025)

Research Advising

Master's Student: Sathvik Bandaru (co-advising)

Presentations

Talks

T1. **Huang, Fan**. *AI-Assisted Research* Invited talk, Prof. Thai Le's lab, Indiana University Bloomington. Apr. 2026.

T2. **Huang, Fan**. *LLM Belief Atlas* Poster presentation at InfoVis 2026 (Boston). 2026.

T3. **Huang, Fan**. *LLM Moral Reasoning Trajectory Analysis* Invited talk, Prof. Han Xue's lab, Donghua University, Shanghai. June 2026.

T4. **Huang, Fan**. *Understanding Moral Reasoning Trajectories in Large Language Models: Toward Probing-Based Explainability* Parallel talk at IC2S2 2026 (Burlington, Vermont). 2026.

T5. **Huang, Fan**. *Vulnerability of LLMs' Stated Belief? LLMs Belief Resistance Check Through Strategic Persuasive Conversation Interventions* Presentation at MSLD 2026 (UIUC). Mar. 2026.

T6. **Huang, Fan**. *ChatGPT Rates Like Human: Towards a Better Alignment of Text Explanation Quality Assessments* Oral presentation at MSLD 2024 (Ann Arbor, Michigan, USA). 2024.

T7. **Huang, Fan**. *Is ChatGPT Better than Human Annotators? & Chain of Explanation* Poster presentation at The Web Conference (WWW) 2023 (Austin, Texas, USA). 2023.

T8. **Huang, Fan**. *Natural Language Explanation: From XAI to Human-AI Interaction* YY Group, CNetS, Indiana University Bloomington. 2023.

Workshops

WK1. **Huang, Fan**. *Participant*, *Summer Institute of Computational Social Science (SICSS)* NUS, Singapore. 2022.

WK2. **Huang, Fan**. *Co-organizer*, *Workshop "Introduction to NLP"* IEEE NTU Student Branch, NTU, Singapore. Oct. 2021.

Academic Service

Program Committee Member

2026	WISE 2026
2026	IC2S2 2026
2026	WWW 2026: Web4Good
2026	7th CySoc Workshop at ICWSM 2026
2025	6th CySoc Workshop at ICWSM 2025
2025	WebSci 2025

Conference Reviewer

2026	NeurIPS 2026
2026	KDD 2026
2026	COLM 2026
2026	IUI 2026
2026	ACL Rolling Review (ARR) 2026
2026	ICWSM 2026
2025	IMX 2025 (ACM International Conference on Interactive Media Experiences)
2025	ACL Rolling Review (ARR) 2025
2025	CHI 2025
2025	ICWSM 2025
2025	IDC 2025

2022 WSDM 2022

Journal Reviewer

2025 PeerJ Computer Science
2024–25 Online Social Networks and Media
2024 Information Processing & Management

Organized Activities / Volunteer

2026 IC2S2 2026: Science of Science (AI) Parallel Talk Session, Session Chair
2025 SODA Lab Research Talk: 'Multimodal NLP Research Insights', Dr. Henry Wei Han, Organizer (Nov 2025)
2024– LLM Innovators Hub @ Soda Lab, Organizer (since Oct 2024)
2023–24 CNetS @ Luddy School Gathering Event, Organizer (Aug 2023–Oct 2024)
2023 ISSI 2023, Student Volunteer

Technical Skills

Programming Python, R, SQL, Java
ML/NLP PyTorch, Hugging Face Transformers, LangChain, OpenAI API, scikit-learn, spaCy, NLTK
LLM Tools Fine-tuning (LoRA, PEFT), Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Evaluation
Data & Viz Pandas, NumPy, Matplotlib, Seaborn, NetworkX
Platforms Linux/Unix, Git, Docker, HPC/Slurm, Jupyter

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